



Bayesian inference for the dynamic properties of long-span bridges under vortex-induced vibration with Scanlan's model and dense optical flow scheme

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ABSTRACT

The dynamic properties of long-span bridges with flexible and light-weight features are of critical importance in safety assessment and response prediction. The primary concern of this study was to infer the dynamic characteristics of long-span bridges subjected to vortex-induced vibration (VIV) by reasonably accommodating the fluid–structure interactions during VIV and the various uncertainties arising in the dynamic characterization of the structures due to measurement noise and modeling errors. By taking the advantage of advanced image processing techniques, the Farnebäck dense optical flow method was used to process the vibration video to extract the displacement time history. In contrast to conventional operational modal analysis (OMA) approaches, which are usually based on the white noise assumption, a new frequency-domain Bayesian method is proposed to make a statistical inference for the dynamic characteristics of the bridge based on the power spectral density of the VIV measurements and the Scanlan's empirical VIV model for bridge decks. A fast computation scheme is also proposed to achieve the posterior uncertainties of the modal frequency of the bridge, the structural damping of the bridge, and the total damping of the VIV system. The efficiency and the accuracy of the proposed algorithms have been verified with the VIV motion of a super long-span suspension bridge. Comparison with those identified using different OMA approaches in the time domain and the frequency domain is also presented.

1. Introduction

Rapid technological advancement and economic development have allowed the construction of a significant number of large-scale bridges with long spans and super long spans over the past few decades [1,2]. With the increase of scale, modern bridges have become increasingly flexible, complicated, and vulnerable to external excitation [3,4]. As a result, the problem of vibrations affecting driving safety and comfort has been increasingly prominent for long-span bridges [5,6]. Flexible and light-weight bridges are extremely

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sensitive to wind excitations that can cause aggressive motion, thereby leading to progressive damage, fatigue accumulation, and structural collapse [7]. For example, the shaking of a suspension bridge in China in 2020 has impacted the traffic across the Pearl River Delta [8].

Dynamic properties of long-span bridges such as natural frequencies, damping ratios, and mode shapes provide a starting point for quantitatively characterizing the in-service structural characteristics and dynamic behaviors [9,10]. More accurate structural dynamic properties lead to the more accurate understanding of the performance of the structures [11,12]. In principle, the modal parameters can be identified by using free vibration testing [13], forced vibration testing [14], and ambient vibration testing [15]. For long-span bridges, one can rely upon ambient vibration by making full use of a variety of natural excitation sources such as wind, traffic, or ocean waves [16]. Compared to free vibration or forced vibration testing, the obvious advantage of ambient vibration testing is its economy and convenience in implementation. Furthermore, the dynamic properties identified in an ambient vibration situation can represent the real operating conditions of a structure during its daily use [9]. In theory, modal analysis can be performed in the time domain, frequency domain, or time–frequency domain. Temporal signals directly processed in the time domain for OMA mainly include stochastic subspace identification (SSI) and its reference-based generalization (SSI/ref) [17], the random decrement Ibrahim time domain (RD-ITD) method [18]. Frequency-domain methods make use of the matrix of output spectral densities. There are four main types of frequency-domain methods, including the frequency domain decomposition (FDD) method [19], the transmissibility function method [20,21], the poly-reference least-squares complex frequency domain (PolyMAX) method [22], and Bayesian operational modal identification [23–25].

A significant number of field tests have been performed on long-span bridges experiencing ambient vibration over the past few decades [26] by using various OMA approaches. However, studies reporting on the investigation of the dynamic characteristics of long-span bridges when the bridges were subjected to vortex-induced vibration (VIV) have been rare. VIV is a large amplitude oscillation phenomenon that often occurs at low and moderate wind speed [27,28]. VIV can cause a usability problem and even cause fatigue failures for some components. With the ever-present growth of span lengths, bridges are becoming more flexible and thereby more susceptible to VIV [29]. Careful attention should be paid to assess the dynamic characteristics of bridges experiencing VIV.

Some difficulties may arise in identifying the dynamic characteristics of long-span bridges subjected to VIV. Generally, no testing data are released to the public, and only short videos taken by citizens passing through are available when VIV occurs. However, although conventional OMA methods have been applied to analyze dynamic properties [30], conventional OMA theories cannot account for the effect of fluid–structure interaction with additional aeroelastic stiffness and damping acting on a deck during VIV [31]. Furthermore, it is highly nontrivial to identify VIV forces that do not satisfy the mathematical model, such as the white noise assumption commonly made in conventional OMA. Since the use of video sequences for extracting vibration measurement has increasingly become feasible, the first difficulty can be well addressed by utilizing computer vision techniques [32–35]. With the development of wind engineering treating a bridge and the surrounding flow field as a unit system with nonlinear damping properties, the second difficulty can be addressed by using some models that properly account for fluid–structure interactions [31]. Therefore, the aim of this study is to propose a new approach to statistically infer the dynamic characteristics of a bridge experiencing VIV by incorporating the Farnebäck dense optical flow method [36,37] and the Scanlan's empirical model [38]. The main contributions of this study are threefold:

- Combined with the Farnebäck dense optical flow algorithm [36], the VIV displacement of the bridge was extracted by processing camera-based video. The optical flow techniques making full use of the velocity field resulting from an intensity change on the image plane were powerful. The dense optical flow presenting the flow vectors of the entire frame (all pixels) up to one flow vector per pixel could give higher accuracy compared with sparse optical flow. This was a point-by-point method for deriving the moving direction and velocity of targets from the analysis of the displacements of these targets in two successive video images.
- Based on the displacements extracted using the Farnebäck dense optical flow algorithm, a frequency-domain Bayesian scheme was formulated to make a statistical inference for the dynamic characteristics of the bridge based on the power spectral density (PSD) of the VIV measurements and the single-degree-of-freedom (SDOF) empirical VIV model for bridge decks [38]. A fast computation scheme was also proposed to achieve the posterior uncertainties of the natural frequency of the bridge as well as the vortex shedding, the structural damping of the bridge, and the total damping of the VIV system. As a result, the fluid–structure interactions during VIV and various uncertainties arising in the dynamic testing of the structures due to measurement noise and modeling errors could be well accounted for.
- The efficiency and the accuracy of the proposed algorithms were verified using the VIV motion of a super long-span suspension bridge. Different OMA approaches in the time domain, such as the SSI, and in the frequency domain, such as PolyMax, Bayesian method and Power Spectral Density Transmissibility (PSDT), were also used to determine the dynamic properties of the bridge. The outcomes were compared by evaluating the performance and reliability of the proposed method. The case study could provide an important supplement to the structural performance assessment of long-span bridges experiencing VIV.

The paper is organized as follows. The rationale of the displacement extraction based on the Farnebäck dense optical flow method is presented in Section 2. Section 3 presents the frequency-domain Bayesian formulation for identifying the dynamic characteristics of a bridge based on the VIV measurements. The results of the Farnebäck dense optical flow method and the frequency domain Bayesian formulation approach were verified using the VIV measurements of the suspension bridge, as described in Section 4, accompanied by comparisons and discussions.

2. Farneback optical flow-based VIV measurements

Since large deformation occurs when a bridge is subject to VIV, an optical flow method can be used to calculate the variation of the optical intensity of a selected region of interest (ROI) in a girder image sequence. The obtained optical flow vector provides the direction of vibration for the ROI on the girder segment, which can then be used to estimate the displacement of the ROI on the image plane. There are three key steps in video processing, i.e., the detection of moving objects, the tracking of such objects from frame to frame, and the analysis of object tracks to recognize the behavior [32].

2.1. Fundamentals of optical flow and the governing equation

Optical flow is the instantaneous velocity of the pixel movement of a moving object on the imaging plane. The fundamental rationale of the optical flow method is to discover the previous frame and the current frame correspondence utilizing the change of the pixel in the image sequence as well as the correlation between adjacent frames. As a result, one can compute the motion of the objects between adjacent frames in the time domain.

The fundamental hypothesis of optical flow algorithms is the conservation of voxels and image pixels, which assumes that the color of an object does not change significantly in the previous two frames [32]. It is worth noting that ambient light usually does not change drastically from the one frame to the next, indicating that the assumption is usually satisfied in video processing. Based on these assumptions, the image constraint equation can be derived. By imposing different additional conditions on the image constraint equation, one can further develop various types of algorithms.

As is shown in Fig. 1, we assumed that each small grid represented a pixel. The pixel P in the image represents a tracing target point, which moves to pixel P' within the time interval of two successive images denoted by Δt . By assuming that $I(x_p, y_p, t)$ denotes the gray value of $P(x_p, y_p)$ at time t , the gray value of $P'(x_p + \Delta x, y_p + \Delta y)$ at time $t + \Delta t$ can be expressed as $I(x_p + \Delta x, y_p + \Delta y, t + \Delta t)$. According to the aforementioned grayscale invariant hypothesis, one has [32]:

$$I(x_p, y_p, t) = I(x_p + \Delta x, y_p + \Delta y, t + \Delta t) \quad (1)$$

Using the Taylor series expansion, Eq. (1) can be expanded as

$$I(x_p + \Delta x, y_p + \Delta y, t + \Delta t) = I(x_p, y_p, t) + \frac{\partial I}{\partial x} \Delta x + \frac{\partial I}{\partial y} \Delta y + \frac{\partial I}{\partial t} \Delta t + \varepsilon \quad (2)$$

By substituting Eq. (1) into Eq. (2) and ignoring the higher order terms in Eq. (2), one can obtain

$$\frac{\partial I}{\partial x} \frac{\Delta x}{\Delta t} + \frac{\partial I}{\partial y} \frac{\Delta y}{\Delta t} + \frac{\partial I}{\partial t} = 0 \quad (3)$$

Eq. (3) is the fundamental equation of the optical flow method. The Farneback optical flow method used in this study is one of most accurate object motion estimation methods based on video processing techniques.

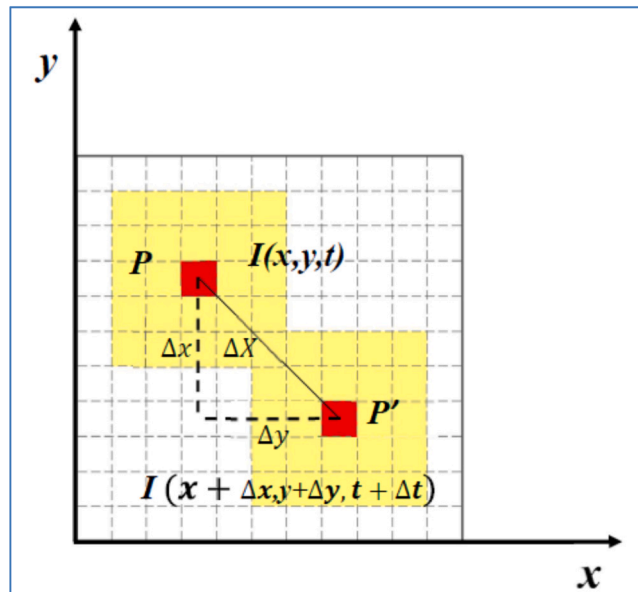


Fig. 1. Schematic diagram of gray scale in optical flow problem.

2.2. Farnebäck optical flow method

In addition to the gray level invariant hypothesis, the Farnebäck dense optical flow method [36,37] has an additional hypothesis of velocity smoothing constraints in that the adjacent points in the image plane move in a similar way. For a small image neighborhood, the gray level function $I(\mathbf{X})$ of a pixel $\mathbf{X} = (x, y)^T$ can be expressed with a quadratic polynomial as follows:

$$I(\mathbf{X}) = \mathbf{X}^T \mathbf{A} \mathbf{X} + \mathbf{b}^T \mathbf{X} + c \quad (4)$$

where $\mathbf{A}$ is a symmetric matrix; $\mathbf{b}$ is a vector; and c is a scalar. At time t , it is assumed that the gray value of the pixel $\mathbf{X}$ is denoted by

$$I_t(\mathbf{X}) = \mathbf{X}^T \mathbf{A}_t \mathbf{X} + \mathbf{b}_t^T \mathbf{X} + c_t \quad (5)$$

After a short time interval Δt , the displacement of $\mathbf{X}$ is denoted by $\Delta \mathbf{X}$ ($\Delta x, \Delta y$), so the gray value of $(\mathbf{X} + \Delta \mathbf{X})$ is given by

$$\begin{aligned} I_{t+\Delta t}(\mathbf{X} + \Delta \mathbf{X}) &= (\mathbf{X} + \Delta \mathbf{X})^T \mathbf{A}_{t+\Delta t} (\mathbf{X} + \Delta \mathbf{X}) + \mathbf{b}_{t+\Delta t}^T (\mathbf{X} + \Delta \mathbf{X}) + c_{t+\Delta t} \\ &= \mathbf{X}^T \mathbf{A}_{t+\Delta t} \mathbf{X} + (\mathbf{b}_{t+\Delta t} + 2\mathbf{A}_{t+\Delta t} \Delta \mathbf{X})^T \mathbf{X} + \Delta \mathbf{X}^T \mathbf{A}_{t+\Delta t} \Delta \mathbf{X} + \mathbf{b}_{t+\Delta t}^T \Delta \mathbf{X} + c_{t+\Delta t} \end{aligned} \quad (6)$$

According to the gray level invariant hypothesis, the following equation should be satisfied:

$$I_t(\mathbf{X}) = I_{t+\Delta t}(\mathbf{X} + \Delta \mathbf{X}) \quad (7)$$

Substituting Eq. (5) and (6) into Eq. (7) leads to

$$\mathbf{A}_t = \mathbf{A}_{t+\Delta t} \quad (8a)$$

$$\mathbf{b}_t = \mathbf{b}_{t+\Delta t} + 2\mathbf{A}_{t+\Delta t} \Delta \mathbf{X} \quad (8b)$$

$$c_t = c_{t+\Delta t} + \Delta \mathbf{X}^T \mathbf{A}_{t+\Delta t} \Delta \mathbf{X} + \mathbf{b}_{t+\Delta t}^T \Delta \mathbf{X} \quad (8c)$$

If $\mathbf{A}_{t+\Delta t}$ is non-singular, one can solve for the translation $\Delta \mathbf{X}$ from Eq. (8):

$$\Delta \mathbf{X} = -\frac{1}{2} \mathbf{A}_{t+\Delta t}^{-1} (\mathbf{b}_{t+\Delta t} - \mathbf{b}_t) \quad (9)$$

According to Eq. (8a), $\mathbf{A}_{t+\Delta t} = \mathbf{A}_t$. However, in practice, one should settle for the following approximation:

$$\mathbf{A}_M = \frac{1}{2} (\mathbf{A}_t + \mathbf{A}_{t+\Delta t}) \quad (10)$$

After introducing $\Delta \mathbf{b} = -1/2(\mathbf{b}_{t+\Delta t} - \mathbf{b}_t)$ and using $\mathbf{A}_M$ to replace $\mathbf{A}_{t+\Delta t}$, Eq. (9) can be rearranged as

$$\mathbf{A}_M \Delta \mathbf{X} = \Delta \mathbf{b} \quad (11)$$

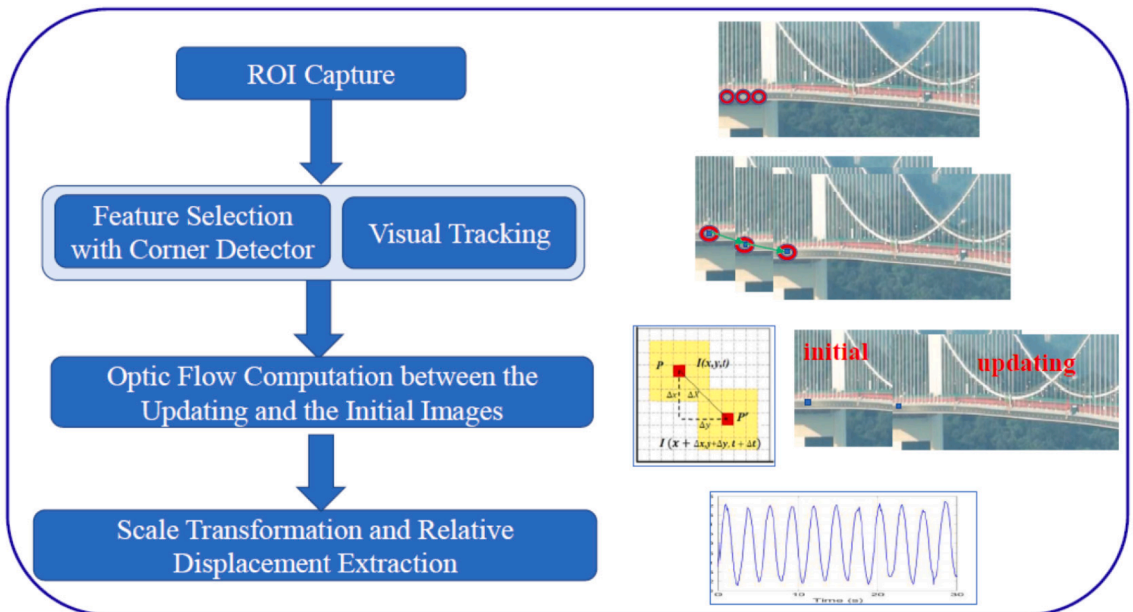


Fig. 2. Flowchart of video processing using the Farnebäck optical flow algorithm.

In principle, Eq. (11) can be solved pointwise, but the results turn out to be noisy. To reduce the influence of noise, the error function $\varepsilon(\mathbf{X})$ on the neighborhood (Ω) of the target pixel is constructed. The displacement ($\Delta\mathbf{X}$) of the target point is determined by minimizing the error function $\varepsilon(\mathbf{X})$, from which the velocity of the pixel is deduced:

$$\varepsilon(\mathbf{X}) = \sum_{\Delta\mathbf{X} \in \Omega} w(\Delta\mathbf{X}) \|\mathbf{A}_M(\mathbf{X} + \Delta\mathbf{X})\Delta\mathbf{X} - \Delta\mathbf{b}(\mathbf{X} + \Delta\mathbf{X})\|^2 \quad (12)$$

where $\|\cdot\|$ denotes the Euclidean norm; and $w(\Delta\mathbf{X})$ denotes the Gauss weighting function in the neighbourhood Ω , which is used to reflect the influence degree of each point in Ω . The closer each pixel to the target pixel in the neighbourhood Ω , the greater the value of the Gauss weighting function. The solution of $\Delta\mathbf{X}$ is obtained as

$$\Delta\mathbf{X} = \left(\sum_{\Delta\mathbf{X} \in \Omega} w(\Delta\mathbf{X}) \mathbf{A}_M^T \mathbf{A}_M \right)^{-1} \sum_{\Delta\mathbf{X} \in \Omega} w(\Delta\mathbf{X}) \mathbf{A}_M^T \Delta\mathbf{b} \quad (13)$$

Eq. (13) indicates that $\mathbf{A}_M^T \mathbf{A}_M$, $\mathbf{A}_M^T \Delta\mathbf{b}$, and $\Delta\mathbf{b}^T \Delta\mathbf{b}$ are computed pointwise, and these are averaged with w before the displacement is computed. The robustness can be further improved if the displacement field can be parameterized according to some motion model. It is straightforward to adopt linear motion models, such as the affine motion model or the eight-parameter model. One can refer to reference [36] for more information on the modeling issue.

2.3. Flowchart of the proposed method

The flowchart of the vision-based displacement measurement based on the Farnebäck dense optical flow method is illustrated in Fig. 2. First, one can artificially capture an initial image of the ROI to ensure that the displacements could be extracted at the expected locations on the girder. Each of the measurement points represented by an ROI is a subset selected from the whole image. Then features are extracted from the ROI based on corner detectors, while the locations of the measurement targets (i.e., the extracted features from the original image) are determined for each successive image through visual tracking. With feature extraction and visual tracking, the optic flow for each feature point from the selected ROI in two successive frames can be obtained according to Farnebäck optical flow method introduced in Section 2.2. Once the optical flow from the previous frame to the current frame is achieved, one can compute the displacements of decrement for different targeted feature points using the scale transformation. Then, the displacement decrement of each selected ROI from two frames is calculated by averaging the decrements for each feature point.

3. Bayesian spectral density approach based on Scanlan's empirical VIV model

3.1. The SDOF empirical VIV model

As a large-amplitude oscillation phenomenon often occurring at low and moderate wind speed, VIV will include additional aeroelastic effects on a deck [31]. A model that can reasonably simulate the fluid–structure interactions during VIV is of critical importance for the robust inference of the dynamic characteristics. The SDOF empirical VIV model of a bridge deck [38] shown in Fig. 3 first proposed by Scanlan was used in this study. This model treated the bridge and the surrounding flow field as a unit system with nonlinear damping properties. The SDOF empirical model can reasonably reflect the energy transfer from the flow field to the deck during VIV and they are easy to implement. It has been widely reported that the Scanlan's SDOF empirical model is a good model to well predict the prototype response of a bridge [31].

The SDOF empirical model had the following equation of motion [38]:

$$m(\ddot{v}(t) + 2\zeta_n \omega_n \dot{v}(t) + \omega_n^2 v(t)) = \frac{1}{2} \rho U^2 B \left(KH_1^* \frac{\dot{v}(t)}{U} - C_L(K) \sin(\omega_s t) \right) \quad (14)$$

where $\ddot{v}(t)$, $\dot{v}(t)$ and $v(t)$ denote the acceleration, velocity, and displacement of the bridge deck during VIV; m denotes the mass of a unit length of the bridge deck; ρ, B, D , and U denote the air density, deck width, deck height and wind speed; ζ_n denotes the structural

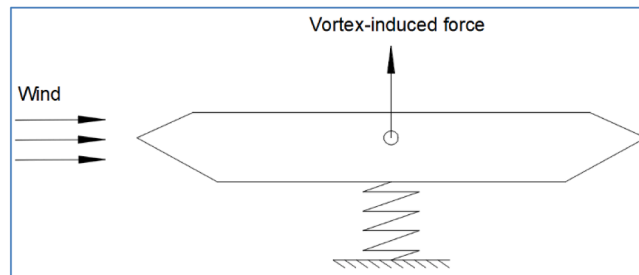


Fig. 3. Schematic diagram of the SDOF model for vertical VIV of the bridge deck section model.

damping; ω_n and ω_s denote the structural natural frequency and the vortex shedding frequency. In vortex-induced resonance, the bridge vibration is usually dominated by a vertical bending or torsional mode, and the frequency of vibration is close to the structural natural frequency. Therefore, ω_s will be replaced by ω_n in the following derivation. The parameters K, C_L , and H_1^* denote the reduced vibration frequency, the magnitude of the forcing term, and one flutter derivatives, which are given as follows [38]:

$$K = \frac{\omega_n B}{U} \quad (15a)$$

$$H_1^* = \frac{4m}{\rho B^2} (\zeta_n - \zeta) \quad (15b)$$

$$C_L = \frac{4m\zeta K^2}{\rho B^2} \frac{A_s}{B} \quad (15c)$$

where A_s is the approximate steady-state VIV amplitude and ζ is the total damping of the VIV system.

It is worth noting that the above equation is suitable for bridge deck section model. The vortex-induced amplitude of SDOF model can be converted to that of the full bridge by making modifications for the above equation [39]:

$$\ddot{q}(t) + 2\zeta_n \omega_n \dot{q}(t) + \omega_n^2 q(t) = \frac{1}{2} m_{eq}^{-1} \rho U^2 B K H_1^* \frac{\dot{q}(t)}{U} - \frac{1}{2} m_{eq}^{-1} \frac{\int_0^L \varphi(x) dx}{\int_0^L \varphi^2(x) dx} \rho U^2 B (C_L(K) \sin(\omega_n t)) \quad (16)$$

where $q(t)$ denotes the modal coordinate; $\varphi(x)$ denotes the bending mode shape of the girder of the full bridge along the longitudinal direction; L denotes the length of the girder of the full bridge; m_{eq} denotes the equivalent mass of the full bridge given by:

$$m_{eq} = \tilde{M}_v / \int_0^L \varphi^2(x) dx \quad (17)$$

where $\tilde{M}_v$ denotes modal mass and it is equal to unit if the mode shape is mass-normalized.

Since the displacement at spanwise point x_i in a single mode is given by $v(t) = \varphi(x_i)q(t)$, the equivalent equation of motion can be derived based on Eq. (16):

$$\ddot{v}(t) + 2\zeta_n \omega_n \dot{v}(t) + \omega_n^2 v(t) = \frac{1}{2} m_{eq}^{-1} \rho U^2 B K H_1^* \frac{\dot{v}(t)}{U} - \frac{1}{2} m_{eq}^{-1} \beta \rho U^2 B (C_L(K) \sin(\omega_n t)) \quad (18)$$

where $\beta = \varphi(x_i) \int_0^L \varphi(x) dx / \int_0^L \varphi^2(x) dx$; and $\varphi(x_i)$ denotes the mode shape at spanwise point x_i . Taking the Fourier transform for Eq. (18) gives:

$$\left[-\omega_k^2 + i\omega_k \left(2\zeta_n \omega_n - \frac{1}{2} m_{eq}^{-1} \rho U B K H_1^* \right) + \omega_n^2 \right] V_k = -\frac{1}{2} m_{eq}^{-1} \beta \rho U^2 B C_L(K) \mathfrak{S}_k(\sin(\omega_n t)) \quad (19)$$

where V_k denotes the frequency response of $v(t)$ at ω_k ; and $\mathfrak{S}_k(\cdot)$ denotes the Fourier transform coefficient of $\sin(\omega_n t)$ at ω_k . Substituting Eq. (15) into Eq. (19) results in:

$$\left[-\omega_k^2 + i\omega_k \left(2\zeta_n \omega_n - 2m_{eq}^{-1} m \omega_n (\zeta_n - \zeta) \right) + \omega_n^2 \right] V_k = -2m m_{eq}^{-1} \beta \zeta A_s \omega_n^2 \mathfrak{S}_k(\sin(\omega_n t)) \quad (20)$$

Therefore, the frequency response V_k is determined to be:

$$V_k = -2\beta m_{eq}^{-1} m \zeta \omega_n^2 A_s \left[-\omega_k^2 + 2i\omega_k \left(\zeta_n \omega_n - m_{eq}^{-1} m \omega_n (\zeta_n - \zeta) \right) + \omega_n^2 \right]^{-1} \mathfrak{S}_k(\sin(\omega_n t)) \quad (21)$$

Consequently, the PSD of V_k is equal to [40,41]:

$$V_k V_k^* = 4\beta^2 m_{eq}^{-2} m^2 \zeta^2 \omega_n^4 A_s^2 \left[(\omega_n^2 - \omega_k^2)^2 + 4\omega_k^2 \left(\zeta_n \omega_n - m_{eq}^{-1} m \omega_n (\zeta_n - \zeta) \right)^2 \right]^{-1} |\mathfrak{S}_k(\sin(\omega_n t))|^2 \quad (22)$$

3.2. Bayesian formulation for inferring the dynamic characteristics

The measured vertical displacement of the bridge deck at the n -th time step $n\Delta t$ ($n = 1, 2, \dots, N$) is denoted by $\tilde{\mathbf{y}}(n) \in \mathbb{R}^{n_0}$, and it is usually assumed to be equal to the sum of model response $\mathbf{v}(n)$ and the prediction error $\mu(n)$, i.e., $\tilde{\mathbf{y}}(n) = \mathbf{v}(n) + \mu(n)$. Its corresponding frequency-domain counterparts at the frequency ω_k are given by [42]:

$$\tilde{\mathbf{Y}}_k = \mathbf{V}_k(\boldsymbol{\theta}) + \mathbf{Y}_k \quad (23)$$

where $\tilde{\mathbf{Y}}_k, \mathbf{V}_k(\boldsymbol{\theta})$, and $\mathbf{Y}_k$ denote the fast Fourier transform (FFT) coefficients of the VIV measurements $\tilde{\mathbf{y}}(n)$, the model response $\mathbf{v}(n)$, and prediction error $\mu(n)$, respectively. $\boldsymbol{\theta}$ denotes the model parameters involved in the SDOF empirical VIV model to be identified.

It has been proved in reference [23,40] that the FFT coefficient $\tilde{Y}_k$ follows a Gaussian distribution as N is large, with the mean $E[Y_k] = 0$ and the variance $S_{Y_k} = E[\tilde{Y}_k \tilde{Y}_k^*]$. Based on Eq. (23), the variance S_{Y_k} is equal to:

$$S_{Y_k} = E[\tilde{Y}_k \tilde{Y}_k^*] = E[V_k(\boldsymbol{\theta}) V_k^*(\boldsymbol{\theta})] + E[Y_k Y_k^*] \quad (24)$$

where $E[V_k(\boldsymbol{\theta}) V_k^*(\boldsymbol{\theta})]$ and $E[Y_k Y_k^*]$ denote the mathematical expectation of the PSD of the model response $V_k(\boldsymbol{\theta})$ and the prediction error Y_k at ω_k , respectively. In real applications, it is usually assumed that the prediction error follows a white noise distribution, i. e., $E[Y_k Y_k^*] = S_\mu$. By combining the formula shown in Eq. (22), Eq. (24) reduces to:

$$S_{Y_k} = 4\beta^2 m_{eq}^{-2} m^2 \zeta^2 \omega_n^4 A_s^2 \left[(\omega_n^2 - \omega_k^2)^2 + 4\omega_k^2 (\zeta_n \omega_n - m_{eq}^{-1} m \omega_n (\zeta_n - \zeta))^2 \right]^{-1} |\Im_k(\sin(\omega_n t))|^2 + S_\mu^2 \quad (25)$$

Given that $\tilde{Y}_k$ follows a Gaussian distribution, the PSD $G_{Y_k}^- = \tilde{Y}_k \tilde{Y}_k^*$ follows a chi-square distribution [23,40]. The probability density function (PDF) of S_{Y_k} is given by [40]:

$$p(G_{Y_k}^-) = c \left| S_{Y_k}^- \right|^{-\frac{1}{2}} e^{\left(-G_{Y_k}^- / S_{Y_k}^- \right)} \quad (26)$$

It is assumed that the PSD set $G_{\wp} = \{G_{Y_k}^-, k = k_1, \dots, k_2\}$ that is formed over the frequency band $\wp \in [k_1 \Delta\omega, k_2 \Delta\omega]$ is used for Bayesian inference. When the number of discrete data points is large, it has been proven that $G_{Y_k}^-$ and $G_{Y_{k'}}^-$ are uncorrelated as $k \neq k'$ [23,40–42]. Based on Bayes' theorem, the posterior PDF of $\boldsymbol{\theta}$ is:

$$p(\boldsymbol{\theta}|\wp) = c_o p(\boldsymbol{\theta}) p(\wp|\boldsymbol{\theta}) \quad (27)$$

In the above equation, the likelihood function $p(G_{\wp}|\boldsymbol{\theta})$ given by:

$$p(G_{\wp}|\boldsymbol{\theta}) = \prod_{k=k_1}^{k_2} c \left| S_{Y_k}^- \right|^{-\frac{1}{2}} e^{\left(-G_{Y_k}^- / S_{Y_k}^- \right)} \quad (28)$$

With the assumption of a non-informative prior distribution, it is convenient to write Eq. (25) in terms of the 'negative log-likelihood function' (NLLF) as:

$$L(\boldsymbol{\theta}) = \frac{1}{2} \sum_{k=k_1}^{k_2} \ln \left| S_{Y_k}^- \right| + \sum_{k=k_1}^{k_2} \frac{G_{Y_k}^-}{S_{Y_k}^-} \quad (29)$$

where $S_{Y_k} = 4\beta^2 (x_i) m_{eq}^{-2} m^2 \zeta^2 \omega_n^4 A_s^2 \left[(\omega_n^2 - \omega_k^2)^2 + 4\omega_k^2 (\zeta_n \omega_n - m_{eq}^{-1} m \omega_n (\zeta_n - \zeta))^2 \right]^{-1} |\Im_k(\sin(\omega_n t))|^2 + S_\mu^2$.

It is worth noting that, in the procedure of Bayesian inference, the parameters to be known in advance include m (the mass of a unit length of the bridge deck), B (deck width), D (deck height), ρ (the air density), U (wind speed), A_s (the approximate steady-state VIV amplitude), and the PSD $G_{Y_k}^-$ estimated from the displacement of the bridge under VIV. The parameters m, B, D can be obtained from the design drawing while ρ, U and A_s can be obtained from the field test. The full set of model parameters to be inferred in the Bayesian framework includes:

$$\boldsymbol{\theta} = \{\omega_n, \zeta_n, \zeta, m_{eq}, \beta, S_\mu\} \quad (30)$$

When the updated (posterior) probability density function (PDF) of the model parameters is very peaked, e.g. the amount of data is large, the updated PDF of the parameter $\boldsymbol{\theta}$ can be well-approximated by a Gaussian PDF with the mean $\hat{\boldsymbol{\theta}}$ and the covariance matrix $\mathbf{H}^{-1}(\hat{\boldsymbol{\theta}})$ [23,24,43]:

- The most probable values (MPVs) of the parameter vector $\hat{\boldsymbol{\theta}}$ can be obtained by minimizing $L(\boldsymbol{\theta})$. In the procedure of optimization for Eq. (30), the initial values of $\boldsymbol{\theta}$ should be estimated in advance [42]. The natural frequency can be estimated using the peak of the PSD; the damping ratio ζ_n and ζ can be estimated as 0.5%; the initial guess of m_{eq} and β can be estimated based on the mode shape computed from the finite element model of the bridge; the initial value of prediction error can be estimated by taking first derivative of the approximation NLLF with respect to S_μ .
- The Hessian matrix $\mathbf{H}(\hat{\boldsymbol{\theta}})$ is deployed as:

$$\mathbf{H}(\hat{\boldsymbol{\theta}}) = \begin{bmatrix} L^{(\omega_n \omega_n)} & L^{(\omega_n \zeta_n)} & L^{(\omega_n \zeta)} & L^{(\omega_n m_{eq})} & L^{(\omega_n \beta)} & L^{(\omega_n S_\mu)} \\ L^{(\zeta_n \omega_n)} & L^{(\zeta_n \zeta_n)} & L^{(\zeta_n \zeta)} & L^{(\zeta_n m_{eq})} & L^{(\zeta_n \beta)} & L^{(\zeta_n S_\mu)} \\ L^{(\zeta \omega_n)} & L^{(\zeta \zeta_n)} & L^{(\zeta \zeta)} & L^{(\zeta m_{eq})} & L^{(\zeta \beta)} & L^{(\zeta S_\mu)} \\ L^{(m_{eq} \omega_n)} & L^{(m_{eq} \zeta_n)} & L^{(m_{eq} \zeta)} & L^{(m_{eq} m_{eq})} & L^{(m_{eq} \beta)} & L^{(m_{eq} S_\mu)} \\ L^{(\beta \omega_n)} & L^{(\beta \zeta_n)} & L^{(\beta \zeta)} & L^{(\beta m_{eq})} & L^{(\beta \beta)} & L^{(\beta S_\mu)} \\ L^{(S_\mu \omega_n)} & L^{(S_\mu \zeta_n)} & L^{(S_\mu \zeta)} & L^{(S_\mu m_{eq})} & L^{(S_\mu \beta)} & L^{(S_\mu S_\mu)} \end{bmatrix} \quad (31)$$

All entries of $\mathbf{H}(\hat{\boldsymbol{\theta}})$ existing in Eq. (31) can be derived analytically similar to reference [42]. The corresponding expressions are presented in the Appendix.

4. Case study of the suspension bridge during VIV

4.1. Introduction of the VIV event of suspension bridge

The suspension bridge under concern is a multiple lane highway suspension bridge that lies in South China. It has important features such as streamlined, low-drag, closed box-girder steel decks exhibiting negative lift in wind conditions. The suspension bridge is located at a wind abundant area of the embouchure of the river. It had a sudden deck vibration that continued intermittently for more than two hours in 2020. The vibration was mainly represented by the vertical vibration of the bridge deck with a maximal amplitude of 31 cm and about 20 cycles per minute, while no obvious torsional vibration was observed. At that time, a hanger was replaced on the site of the bridge deck. For construction safety, water horses with a height of 0.8 m were set up on both sides of the bridge deck as traffic protection equipment [8]. According to the wind speed and frequency locking characteristics when wind induced vibration occurred, the vibration of the bridge deck was justified as a vertical bending VIV attributed to the change of the aerodynamic configuration of the closed steel box girder due to the temporary installation of water horses during the maintenance process [8]. The VIV affected the driving comfort, although it did not endanger the safety of the bridge. VIV was more likely to occur with a low wind speed and low turbulence intensity, when the wind direction was nearly perpendicular to the longitudinal direction of the bridge.

4.2. VIV measurements based on dense optical flow method

To the author's best knowledge, the dynamic responses of the suspension bridge during VIV were not released to the public. However, some videos during the VIV were taken by passengers. With the development of computer vision techniques, the use of video sequences for the extraction of vibration measurement has increasingly become feasible. With permission from the photographer, the video was downloaded and used to verify the methods introduced in the previous sections. The steps of choosing ROIs and the displacements estimation for different targeted locations using the scale transformation were realized by human intervention, while the other steps such as corner detection, visual tracking, and optic flow computation were realized through automatic manner by employing the library of programming functions in OpenCV released to the public. As illustrated in Fig. 4, a number of ROIs on the girder of the bridge were selected as the displacement measurement points. The displacements of the suspension bridge during VIV were extracted by processing this short video using the Farnebäck dense optical flow method. With feature extraction and visual tracking, the optical flow decrement for each feature point from the selected region of interest in two arbitrary consecutive frames was obtained. The sample displacement within a specific time period corresponding to one point is shown in Fig. 5. The sampling frequency was 30 Hz.



Fig. 4. The selected ROI of the bridge deck.

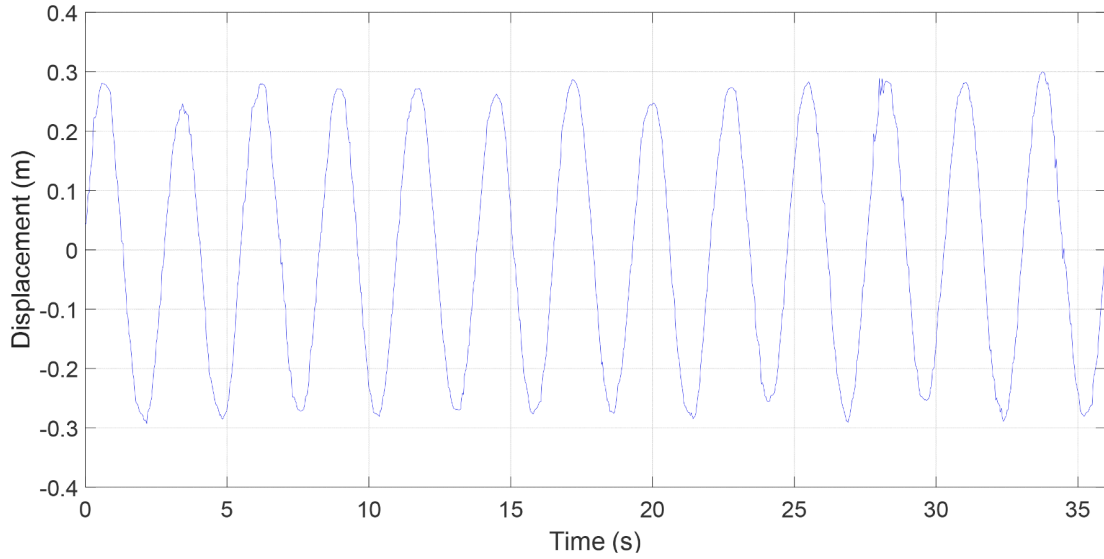


Fig. 5. The displacement corresponding to one location extracted using the dense optical flow method.

4.3. Bayesian inference for the dynamic characteristics

Some critical information for the bridge is provided as follows. The main span of the suspension bridge was $L = 888\text{ m}$; the mass of a unit length of the bridge deck was $m = 23240\text{ kg/m}$; the deck height was $D = 3.012\text{ m}$, and the deck width was $B = 35.6\text{ m}$. The air density was $\rho = 1.25\text{ kg/m}^3$; the wind speed was $U = 8\text{ m/s}$; and the dynamic viscosity was $\lambda = 1.8594 \times 10^{-5}\text{ N} \cdot \text{s/m}^2$. The plot corresponding to the PSD of the displacement of the middle point is shown in Fig. 6. The initial guesses of the structural natural frequency were set to be the peaks of the PSD, while the initial guesses of the structural damping and total damping of the VIV system were assumed to be 0.5%. The PSDs shown in Fig. 6 were used to infer the model parameters θ . Then the MPV of the variables as well as their uncertainties could be estimated based on the procedures introduced in Section 3.2. Table 1 gives a quantitative view of the results in terms of the MPV, the root of the posterior variance, and the posterior coefficient of variation (c.o.v.), which was defined as the standard deviation divided by the MPV. From these results, the following conclusions could be drawn:

- The natural frequency of the bridge identified using the model with consideration of the VIV effect was around 0.368 Hz, which was close to the frequency of the third vertical bending mode identified from previous ambient dynamic testing [44]. Therefore, it was reasonable to justify this event as the VIV of the higher mode. The frequency was also close to the value computed using the empirical formula given in the code of Wind-resistant Design Specification for Highway Bridges in China (JTG/T3360-01-2018, 2018):

$$f_s = 0.198 \times \frac{U}{D} \times \left(1 - \frac{19.7}{Re}\right) = 0.198 \times \frac{8}{3.812} \times \left(1 - \frac{19.7}{4.775 \times 10^8}\right) = 0.415\text{ Hz} \quad (32)$$

where $Re = \frac{\rho UL}{\lambda} = \frac{1.25 \times 8 \times 888}{1.8594 \times 10^{-5}} = 4.775 \times 10^8$. It is worth noting here that D should be equal to 3.812 m, which should be the sum of the bridge deck 3.012 m and the water horse 0.8 m.

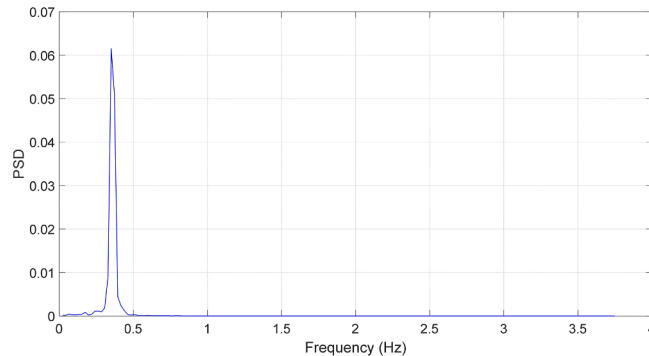


Fig. 6. The PSD of the displacement extracted using dense optical method.

Table 1
Identified dynamic characteristics of the suspension bridge during VIV.

Variables	Values		
	$\hat{\theta}$	σ	$\sigma/\hat{\theta}$ (%)
ω_n (Hz)	0.368	0.005	1.34
ζ_n (%)	0.498	0.093	18.79
ζ (%)	0.539	0.102	18.91
S_μ	7.128	1.786	25.07

- Wind-induced instability of this bridge in erection stage was studied through wind tunnel test of a 1:80 scaled sectional model in smooth flow, and its Reynolds number ranges from 1.5×10^5 to 4.9×10^5 [45]. It has been reported that the identified flutter derivatives of bridge box girders were insensitive to the Reynolds number and turbulence [46]. Given the identified values of ζ_n and ζ in Table 1, H_1^* was computed to be -0.0191 . The results obtained from wind tunnel test and CFD were also reported to be close to zero. As was shown in the figure of Ref. [47], the value of H_1^* obtained from wind tunnel test was estimated around -0.035 and the results estimated from numerical simulation using computational fluid dynamics was around -0.023 . Furthermore, the results agree with the conclusion that closed box-girder steel decks exhibits negative lift [47].
- The posterior c.o.v. values reported in Table 1 reflect the uncertainty associated with the dynamic characteristics given the PSD in the context of the empirical model used. The c.o.v. values of different parameters were of a different order of magnitude. The uncertainties of the structural natural frequency and the structural damping were significantly less than those of the vortex shedding frequency and the total damping of the VIV system, indicating that the latter were identified with lower accuracy. This agrees well with some recent studies [48,49] which also reported that damping has significantly larger uncertainties and scatter range compared with natural frequency. The c.o.v. values of damping ratios with respect to different number of cycles of displacements employed for identification is shown in Fig. 7, which indicates that the uncertainties decrease with the increase of the data length. It would be preferable to include longer measurements if longer videos were available for other case studies.
- Both ζ_n and ζ are identified to be around 0.5%, indicating that the damping ratio in current state is smaller than those in the original design stage and one should take measures to increase the damping effects. It should be pointed out that the vortex-induced force model adopted in this research was only one of the empirical models. If other empirical models [50–52] were to be adopted, the identification results might be different. The comparison and selection of models needs further study in the future.

Based on the displacements extracted using Farnebäck dense optical flow method, OMA was also performed, as described in this section, using different approaches including SSI [17] in the time domain, as well as the PSDT-LSCF approach the [12], PolyMax approach [22], and the fast Bayesian transmissibility approach (B-TOMA) [53] in the frequency domain. The mode shape identified using SSI is shown in Fig. 8. It is worth noting here that all the measurements were fused simultaneously for OMA. The variations of the natural frequencies and damping ratios corresponding to different approaches are compared in Table 2. The outcomes were compared by evaluating the quality and reliability of the results. As seen in Table 2, the natural frequencies identified from all these approaches were close to each other. The identified mode shapes agreed well with each other, with the MAC for each pair approaching over 0.99. However, the damping ratio identified from different approaches indicated much more significant variability.

5. Conclusions

Long-span bridges with flexible and light-weight features are susceptible to wind excitation. Investigation into the dynamic properties of long-span bridges during VIV is of critical importance in safety assessment and response prediction. The aim of this study was to extract the dynamic characteristics of suspension bridges when it was experiencing VIV, based on a camera-based video. To achieve this goal, a computer vision-based method using the Farnebäck dense optical flow technique was used to extract VIV displacements by processing the short camera-based video. Based on the displacements, a new Bayesian inference scheme that reasonably accommodated the fluid–structure interactions during VIV as well as various uncertainties due to the measurement noise and modeling errors were proposed. A fast computation scheme was also proposed to achieve the posterior distribution of the natural frequency of the bridge, the vortex shedding frequency, the structural damping of the bridge, the total damping of the VIV system, and the prediction error between the model and the measurements. The frequency-domain Bayesian formulation proposed in this study could be used to make a statistical inference for dynamic characteristics of the bridge based on the PSD of the VIV measurements and the SDOF empirical VIV model for bridge decks. The results were compared with those identified using different OMA approaches in the time domain and the frequency domain. This study verified that cost-effective and easy-to-use camera-based technologies could support the field inspections of the VIV events of long-span bridges while also eliminating the time-consuming field installation of sensors. More importantly, by conducting modal analysis for the super-long bridge during VIV, new light was also shed on the dynamic performance of this structure with special emphasis on the damping ratio, which could be further increased by structural control measures [54].

CRedit authorship contribution statement

Wang-Ji Yan: Conceptualization, Methodology, Writing – original draft, Funding acquisition. **Zhou-Quan Feng:** Methodology,

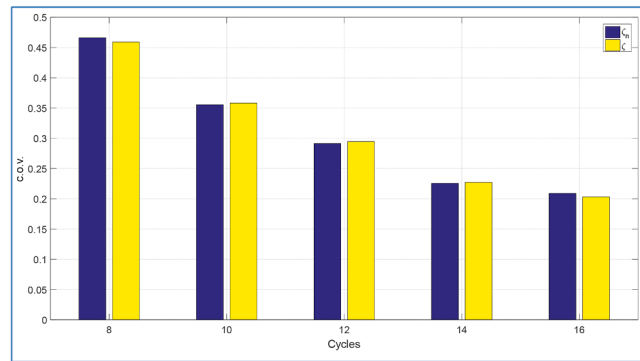


Fig. 7. The variation of c.o.v. values of damping ratio ζ_n and ζ with the increase of the number of cycles.

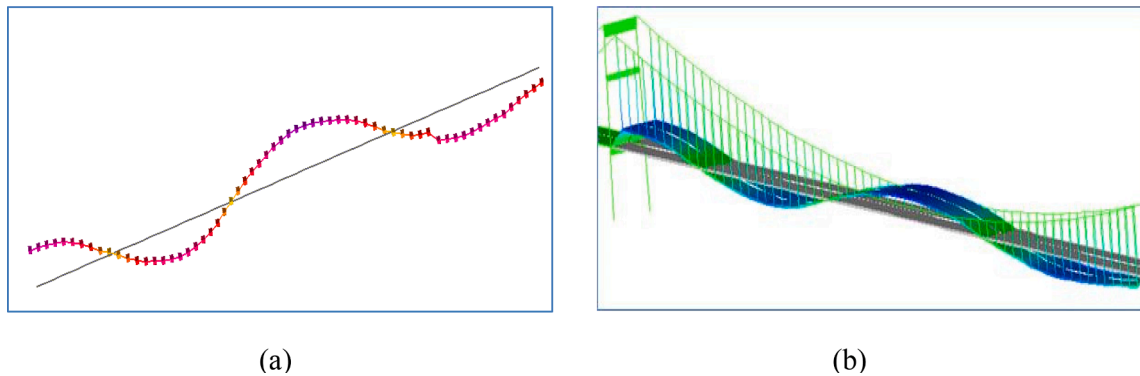


Fig. 8. The mode shape identified using SSI: (a) the 2-D vision; (b) 3-D vision.

Table 2

The natural frequencies and damping ratios identified using different OMA approaches.

Domains	No.	Methods	Frequencies (Hz)	Damping ratio (%)
Time domain	1	SSI [17]	0.361	0.29
Frequency domain	2	PSDT-LSCF [12]	0.355	0.46
	3	PolyMax [22]	0.362	1.78
	4	B-TOMA [53]	0.387	0.53

Funding acquisition, Writing – review & editing. **Wen Yang**: Software of dense optical flow scheme. **Ka-Veng Yuen**: Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

The NLLF of the frequency-domain Bayesian inference scheme shown in [Section 3](#) is given by.

$$L(\boldsymbol{\theta}) = \frac{1}{2} \sum_{k=k_1}^{k_2} \ln S_{Y_k}^{-} + \sum_{k=k_1}^{k_2} \frac{G_{Y_k}^{-}}{S_{Y_k}^{-}} \quad (\text{A1})$$

where $S_{Y_k} = 4\zeta^2 \omega_n^4 A_s^2 \chi_k^{-1} \eta_k + S_\mu^2$ with $\chi_k = (\omega_n^2 - \omega_k^2)^2 + 4\omega_k^2 (\zeta_n \omega_n - m_{eq}^{-1} m \omega_n (\zeta_n - \zeta))^2$ and $\eta_k = \beta^2 m_{eq}^{-2} m^2 \Im(\sin(\omega_n t)) \Im^*(\sin(\omega_k t))$. A parenthesized variable is employed in the superscript to denote a derivative with respect to it.

For any x or y , the derivatives of NLLF $L(\boldsymbol{\theta})$ can be determined as follows:

$$L^{(x)} = \frac{1}{2} \sum_{k=k_1}^{k_2} S_{Y_k}^{-1} S_{Y_k}^{(x)} - \sum_{k=k_1}^{k_2} G_{Y_k}^{-} S_{Y_k}^{-2} S_{Y_k}^{(x)} \quad (\text{A2})$$

The second derivative of $L(\boldsymbol{\theta})$ is given by:

$$L^{(xy)} = \frac{1}{2} \sum_{k=k_1}^{k_2} \left(S_{Y_k}^{(xy)} S_{Y_k}^{-1} - S_{Y_k}^{-2} S_{Y_k}^{(x)} S_{Y_k}^{(y)} \right) - \sum_{k=k_1}^{k_2} G_{Y_k}^{-} \left(S_{Y_k}^{(xy)} S_{Y_k}^{-2} - 2 S_{Y_k}^{-3} S_{Y_k}^{(x)} S_{Y_k}^{(y)} \right) \quad (\text{A3})$$

It is worth noting that the derivative $S_{Y_k}^{-}$ with respect to $\boldsymbol{\theta} = \{\omega_n, \zeta_n, \zeta, m_{eq}, \beta, S_\mu\}$ is given by.

$$S_{Y_k}^{(\omega_n)} = \left(16\omega_n^3 \chi_k^{-1} \eta_k - 4\omega_n^4 \chi_k^{-2} \chi_k^{(\omega_n)} \eta_k + 4\omega_n^4 \chi_k^{-1} \eta_k^{(\omega_n)} \right) \zeta^2 A_s^2 \quad (\text{A4a})$$

$$S_{Y_k}^{(\zeta_n)} = -4\zeta^2 \omega_n^4 A_s^2 \chi_k^{-2} \chi_k^{(\zeta_n)} \eta_k \quad (\text{A4b})$$

$$S_{Y_k}^{(\zeta)} = \left(8\zeta \chi_k^{-1} - 4\zeta^2 \chi_k^{-2} \chi_k^{(\zeta)} \right) \omega_n^4 A_s^2 \eta_k \quad (\text{A4c})$$

$$S_{Y_k}^{(m_{eq})} = -4\zeta^2 \omega_n^4 A_s^2 \chi_k^{-2} \chi_k^{(m_{eq})} \eta_k + 4\zeta^2 \omega_n^4 A_s^2 \chi_k^{-1} \eta_k^{(m_{eq})} \quad (\text{A4d})$$

$$S_{Y_k}^{(\beta)} = 4\zeta^2 \omega_n^4 A_s^2 \chi_k^{-1} \eta_k^{(\beta)} \quad (\text{A4e})$$

$$S_{Y_k}^{(S_\mu)} = 2S_\mu \quad (\text{A4f})$$

Therefore, the derivative of $S_{Y_k}^{(\omega_n)}$ with respect to $\{\omega_n, \zeta_n, \zeta, m_{eq}, \beta, S_\mu\}$ can be further derived as.

$$S_{Y_k}^{(\omega_n \omega_n)} = \zeta^2 A_s^2 \left[\begin{aligned} &48\omega_n^2 \chi_k^{-1} \eta_k - 32\omega_n^3 \chi_k^{-2} \chi_k^{(\omega_n)} \eta_k + 32\omega_n^3 \chi_k^{-1} \eta_k^{(\omega_n)} + 8\omega_n^4 \chi_k^{-3} \left(\chi_k^{(\omega_n)} \right)^2 \eta_k \\ &- 4\omega_n^4 \chi_k^{-2} \chi_k^{(\omega_n \omega_n)} \eta_k - 8\omega_n^4 \chi_k^{-2} \chi_k^{(\omega_n)} \eta_k^{(\omega_n)} + 4\omega_n^4 \chi_k^{-1} \eta_k^{(\omega_n \omega_n)} \end{aligned} \right] \quad (\text{A5a})$$

$$S_{Y_k}^{(\omega_n \zeta_n)} = \left[\left(-16\omega_n^3 \chi_k^{-2} \chi_k^{(\zeta_n)} + 8\omega_n^4 \chi_k^{-3} \chi_k^{(\zeta_n)} \chi_k^{(\omega_n)} - 4\omega_n^4 \chi_k^{-2} \chi_k^{(\zeta_n \omega_n)} \right) \eta_k - 4\omega_n^4 \chi_k^{-2} \chi_k^{(\zeta_n)} \eta_k^{(\omega_n)} \right] \zeta^2 A_s^2 \quad (\text{A5b})$$

$$S_{Y_k}^{(\omega_n \zeta)} = \left(\begin{aligned} &32\omega_n^3 \zeta \chi_k^{-1} - 8\omega_n^4 \zeta \chi_k^{-2} \chi_k^{(\omega_n)} - 16\omega_n^3 \zeta^2 \chi_k^{-2} \chi_k^{(\zeta)} \\ &+ 8\omega_n^4 \zeta^2 \chi_k^{-3} \chi_k^{(\zeta)} \chi_k^{(\omega_n)} - 4\omega_n^4 \zeta^2 \chi_k^{-2} \chi_k^{(\zeta \omega_n)} \end{aligned} \right) A_s^2 \eta_k + \left(8\zeta \chi_k^{-1} - 4\zeta^2 \chi_k^{-2} \chi_k^{(\zeta)} \right) \omega_n^4 A_s^2 \eta_k^{(\zeta \omega_n)} \quad (\text{A5c})$$

$$S_{Y_k}^{(\omega_n m_{eq})} = \left(\begin{aligned} &\left(-16\omega_n^3 \chi_k^{-2} \chi_k^{(m_{eq})} + 8\omega_n^4 \chi_k^{-3} \chi_k^{(m_{eq})} \chi_k^{(\omega_n)} - 4\omega_n^4 \chi_k^{-2} \chi_k^{(m_{eq} \omega_n)} \right) \eta_k - 4\omega_n^4 \chi_k^{-2} \chi_k^{(m_{eq})} \eta_k^{(\omega_n)} \\ &+ 16\omega_n^3 \chi_k^{-1} \eta_k^{(m_{eq})} - 4\omega_n^4 \chi_k^{-2} \chi_k^{(\omega_n)} \eta_k^{(m_{eq})} + 4\omega_n^4 \chi_k^{-1} \eta_k^{(m_{eq} \omega_n)} \end{aligned} \right) \zeta^2 A_s^2 \quad (\text{A5d})$$

$$S_{Y_k}^{(\omega_n \beta)} = \zeta^2 A_s^2 \left(16\omega_n^3 \chi_k^{-1} \eta_k^{(\beta)} - 4\omega_n^4 \chi_k^{-2} \chi_k^{(\omega_n)} \eta_k^{(\beta)} + 4\omega_n^4 \chi_k^{-1} \eta_k^{(\omega_n \beta)} \right) \quad (\text{A5e})$$

$$S_{Y_k}^{(\omega_n S_\mu)} = 0 \quad (\text{A5f})$$

The formulas of $S_{Y_k}^{(\zeta_n x)}$ are given by.

$$S_{Y_k}^{(\zeta_n \zeta_n)} = \left(8\chi_k^{-3} \left(\chi_k^{(\zeta_n)} \right)^2 - 4\chi_k^{-2} \chi_k^{(\zeta_n \zeta_n)} \right) \zeta^2 \omega_n^4 A_s^2 \eta_k \quad (\text{A6a})$$

$$S_{Y_k}^{(\zeta_n \zeta)} = \left[-8\zeta \chi_k^{-2} \chi_k^{(\zeta_n)} - 4\zeta^2 \left(-2\chi_k^{-3} \chi_k^{(\zeta)} \chi_k^{(\zeta_n)} + \chi_k^{-2} \chi_k^{(\zeta_n \zeta)} \right) \right] \omega_n^4 A_s^2 \eta_k \quad (\text{A6b})$$

$$S_{Y_k}^{(\zeta_n m_{eq})} = \left(8\chi_k^{-3} \chi_k^{(m_{eq})} \chi_k^{(\zeta_n)} \eta_k - 4\chi_k^{-2} \chi_k^{(\zeta_n m_{eq})} \eta_k - 4\chi_k^{-2} \chi_k^{(\zeta_n)} \eta_k^{(m_{eq})} \right) \zeta^2 \omega_n^4 A_s^2 \quad (\text{A6c})$$

$$S_{Y_k}^{(\zeta_n \beta)} = -4\zeta^2 \omega_n^4 A_s^2 \chi_k^{-2} \chi_k^{(\zeta_n)} \eta_k^{(\beta)} \quad (\text{A6d})$$

$$S_{Y_k}^{(\zeta_n S_\mu)} = 0 \quad (\text{A6e})$$

The expressions of $S_{Y_k}^{(\zeta x)}$ are given by.

$$S_{Y_k}^{(\zeta \zeta)} = \left(8\chi_k^{-1} - 16\zeta \chi_k^{-2} \chi_k^{(\zeta)} + 8\zeta^2 \chi_k^{-3} \left(\chi_k^{(\zeta)} \right)^2 - 4\zeta^2 \chi_k^{-2} \chi_k^{(\zeta \zeta)} \right) \omega_n^4 A_s^2 \eta_k \quad (\text{A7a})$$

$$S_{Y_k}^{(\zeta m_{eq})} = \left(-8\zeta \chi_k^{-2} \chi_k^{(m_{eq})} + 8\zeta^2 \chi_k^{-3} \chi_k^{(m_{eq})} \chi_k^{(\zeta)} - 4\zeta^2 \chi_k^{-2} \chi_k^{(\zeta m_{eq})} \right) \omega_n^4 A_s^2 \eta_k + \left(8\zeta \chi_k^{-1} - 4\zeta^2 \chi_k^{-2} \chi_k^{(\zeta)} \right) \omega_n^4 A_s^2 \eta_k^{(m_{eq})} \quad (\text{A7b})$$

$$S_{Y_k}^{(\zeta \beta)} = \left(8\zeta \chi_k^{-1} - 4\zeta^2 \chi_k^{-2} \chi_k^{(\zeta)} \right) \omega_n^4 A_s^2 \eta_k^{(\beta)} \quad (\text{A7c})$$

$$S_{Y_k}^{(\zeta S_\mu)} = 0 \quad (\text{A7d})$$

The expressions of $S_{Y_k}^{(m_{eq} m_{eq})}$, $S_{Y_k}^{(m_{eq} \beta)}$, $S_{Y_k}^{(m_{eq} S_\mu)}$, $S_{Y_k}^{(\beta \beta)}$, $S_{Y_k}^{(\beta S_\mu)}$ and $S_{Y_k}^{(S_\mu S_\mu)}$ are given by:

$$S_{Y_k}^{(m_{eq} m_{eq})} = \left[8\chi_k^{-3} \left(\chi_k^{(m_{eq})} \right)^2 \eta_k - 4\chi_k^{-2} \chi_k^{(m_{eq} m_{eq})} \eta_k - 8\chi_k^{-2} \chi_k^{(m_{eq})} \eta_k^{(m_{eq})} + 4\chi_k^{-1} \eta_k^{(m_{eq} m_{eq})} \right] \zeta^2 \omega_n^4 A_s^2 \quad (\text{A8a})$$

$$S_{Y_k}^{(m_{eq} \beta)} = \left[-4\chi_k^{-2} \chi_k^{(m_{eq})} \eta_k^{(\beta)} + 4\chi_k^{-1} \eta_k^{(m_{eq} \beta)} \right] \zeta^2 \omega_n^4 A_s^2 \quad (\text{A8b})$$

$$S_{Y_k}^{(m_{eq} S_\mu)} = 0 \quad (\text{A8c})$$

$$S_{Y_k}^{(\beta \beta)} = 4\zeta^2 \omega_n^4 A_s^2 \chi_k^{-1} \eta_k^{(\beta \beta)} \quad (\text{A8d})$$

$$S_{Y_k}^{(\beta S_\mu)} = 0 \quad (\text{A8e})$$

$$S_{Y_k}^{(S_\mu S_\mu)} = 2 \quad (\text{A8f})$$

In the above equations, $\chi_k^{(x)}$ and $\chi_k^{(xy)}$ should be computed. $\chi_k^{(x)}$ is shown as follows:

$$\chi_k^{(\omega_n)} = 4(\omega_n^2 - \omega_k^2) \omega_n + 8\omega_k^2 \omega_n \left(\zeta_n - m_{eq}^{-1} m(\zeta_n - \zeta) \right)^2 \quad (\text{A9a})$$

$$\chi_k^{(\zeta_n)} = 8\omega_k^2 \left(\zeta_n \omega_n - m_{eq}^{-1} m \omega_n (\zeta_n - \zeta) \right) (\omega_n - m_{eq}^{-1} m \omega_n) \quad (\text{A9b})$$

$$\chi_k^{(\zeta)} = 8m_{eq}^{-1} m \omega_k^2 \omega_n^2 \left(\zeta_n - m_{eq}^{-1} m (\zeta_n - \zeta) \right) \quad (\text{A9c})$$

$$\chi_k^{(m_{eq})} = 8\omega_k^2 m m_{eq}^{-2} \omega_n^2 \left(\zeta_n - m_{eq}^{-1} m (\zeta_n - \zeta) \right) (\zeta_n - \zeta) \quad (\text{A9d})$$

By further processing Eq. (A10), $\chi^{(xy)}$ can be derived as.

$$\chi_k^{(\omega_n \omega_n)} = 12\omega_n^2 - 4\omega_k^2 + 8\omega_k^2 \left(\zeta_n - m_{eq}^{-1} m (\zeta_n - \zeta) \right)^2 \quad (\text{A10a})$$

$$\chi_k^{(\omega_n \zeta_n)} = 16\omega_k^2 \omega_n (\zeta_n - m_{eq}^{-1} m (\zeta_n - \zeta)) (1 - m_{eq}^{-1} m) \quad (A10b)$$

$$\chi_k^{(\omega_n \zeta)} = 16m_{eq}^{-1} m \omega_k^2 \omega_n (\zeta_n - m_{eq}^{-1} m (\zeta_n - \zeta)) \quad (A10c)$$

$$\chi_k^{(\omega_n m_{eq})} = 16m_{eq}^{-2} m \omega_k^2 \omega_n (\zeta_n - m_{eq}^{-1} m (\zeta_n - \zeta)) (\zeta_n - \zeta) \quad (A10d)$$

$$\chi_k^{(\zeta_n \zeta_n)} = 8\omega_k^2 \omega_n^2 (1 - m_{eq}^{-1} m)^2 \quad (A10e)$$

$$\chi_k^{(\zeta_n \zeta)} = 8\omega_k^2 m_{eq}^{-1} m \omega_n^2 (1 - m_{eq}^{-1} m) \quad (A10f)$$

$$\chi_k^{(\zeta_n m_{eq})} = 8\omega_k^2 m m_{eq}^{-2} \omega_n^2 [(\zeta_n - \zeta)(1 - m m_{eq}^{-1}) + (\zeta_n - m m_{eq}^{-1} (\zeta_n - \zeta))] \quad (A10g)$$

$$\chi_k^{(\zeta \zeta)} = 8(m_{eq}^{-1} m \omega_k \omega_n)^2 \quad (A10h)$$

$$\chi_k^{(\zeta m_{eq})} = 8\omega_k^2 \omega_n^2 m_{eq}^{-3} (\zeta_n - \zeta) - 8\omega_k^2 \omega_n^2 m m_{eq}^{-2} (\zeta_n - m m_{eq}^{-1} (\zeta_n - \zeta)) \quad (A10i)$$

$$\chi_k^{(m_{eq} m_{eq})} = 8\omega_k^2 m^2 m_{eq}^{-4} \omega_n^2 (\zeta_n - \zeta)^2 - 16\omega_k^2 m m_{eq}^{-3} \omega_n^2 (\zeta_n - m m_{eq}^{-1} (\zeta_n - \zeta)) (\zeta_n - \zeta) \quad (A10j)$$

As is seen from (A4)-(A8), the derivatives $\eta_k^{(\omega_n)}, \eta_k^{(m_{eq})}, \eta_k^{(\beta)}, \eta_k^{(\omega_n \omega_n)}, \eta_k^{(\omega_n m_{eq})}, \eta_k^{(\omega_n \beta)}, \eta_k^{(m_{eq} m_{eq})}, \eta_k^{(m_{eq} \beta)}$ and $\eta_k^{(\beta \beta)}$ are required, and they are derived as follows:

$$\eta_k^{(\omega_n)} = (\beta m_{eq}^{-1} m)^2 [\Im(tc \cos(\omega_n t)) \Im^*(\sin(\omega_n t)) + \Im(\sin(\omega_n t)) \Im^*(tc \cos(\omega_n t))] \quad (A11a)$$

$$\eta_k^{(m_{eq})} = -2\beta^2 m_{eq}^{-3} m^2 \Im(\sin(\omega_n t)) \Im^*(\sin(\omega_n t)) \quad (A11b)$$

$$\eta_k^{(\beta)} = 2\beta m_{eq}^{-2} m^2 \Im(\sin(\omega_n t)) \Im^*(\sin(\omega_n t)) \quad (A11c)$$

$$\eta_k^{(\omega_n \omega_n)} = (\beta m_{eq}^{-1} m)^2 \left[-\Im(t^2 \sin(\omega_n t)) \Im^*(\sin(\omega_n t)) + 2\Im(tc \cos(\omega_n t)) \Im^*(tc \cos(\omega_n t)) - \Im(\sin(\omega_n t)) \Im^*(t^2 \sin(\omega_n t)) \right] \quad (A11d)$$

$$\eta_k^{(\omega_n m_{eq})} = -2\beta^2 m_{eq}^{-3} m^2 [\Im(tc \cos(\omega_n t)) \Im^*(\sin(\omega_n t)) + \Im(\sin(\omega_n t)) \Im^*(tc \cos(\omega_n t))] \quad (A11e)$$

$$\eta_k^{(\omega_n \beta)} = 2\beta (m_{eq}^{-1} m)^2 [\Im(tc \cos(\omega_n t)) \Im^*(\sin(\omega_n t)) + \Im(\sin(\omega_n t)) \Im^*(tc \cos(\omega_n t))] \quad (A11f)$$

$$\eta_k^{(m_{eq} m_{eq})} = 6\beta^2 m^2 m_{eq}^{-4} \Im(\sin(\omega_n t)) \Im^*(\sin(\omega_n t)) \quad (A11g)$$

$$\eta_k^{(m_{eq} \beta)} = -4\beta m_{eq}^{-3} m^2 \Im(\sin(\omega_n t)) \Im^*(\sin(\omega_n t)) \quad (A11h)$$

$$\eta_k^{(\beta \beta)} = 2m_{eq}^{-2} m^2 \Im(\sin(\omega_n t)) \Im^*(\sin(\omega_n t)) \quad (A11i)$$

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